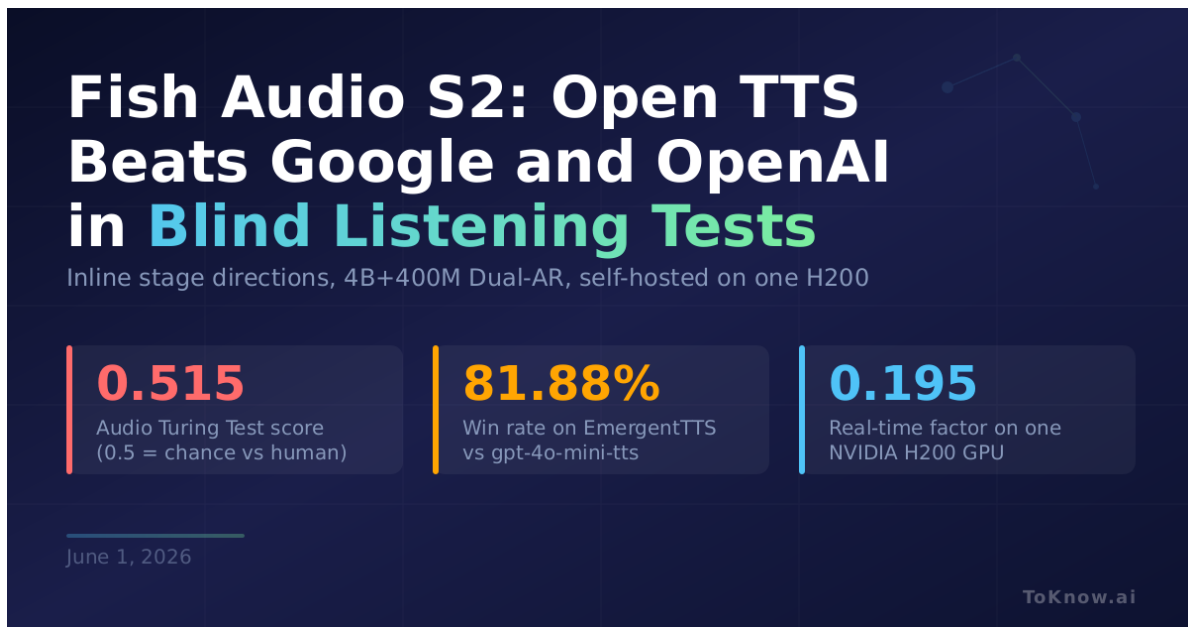


Fish Audio S2: Open-Source Text-to-Speech Beats Google and OpenAI in Blind Listening Tests

Kabui, Charles

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Fish Audio released [S2](#), a text-to-speech system that reads free-form stage directions written inline with the words. Drop [whisper in small voice] or [professional broadcast

tone] next to any phrase and the model steers prosody and emotion to match. Its dual-autoregressive design pairs a 4-billion-parameter model along the time axis with a 400-million-parameter model that fills in acoustic detail at each step. Trained on over 10 million hours of audio in around 50 languages, S2 scored 0.515 on the [Audio Turing Test](#), where 0.5 means listeners label synthetic speech as human about half the time. It also posted the lowest word error rate on the [Seed-TTS Eval](#) benchmark in both Chinese (0.54%) and English (0.99%), beating Seed-TTS, MiniMax Speech-02, and Qwen3-TTS.

High-quality emotional TTS has been a paid API, billed per character. S2 ships model weights, fine-tuning code, and an [SGLang inference engine](#) together, reaching a real-time factor of 0.195 with around 100 milliseconds to first audio on a single NVIDIA H200. A podcaster or game studio can self-host the same quality. The catch: weights are under a Fish Audio research license, not Apache or MIT, so commercial deployment needs a separate agreement.

An open release now leads a public, blind listening benchmark against closed systems from Google and OpenAI. On [EmergentTTS-Eval](#), S2 wins 81.88% of comparisons against a gpt-4o-mini-tts baseline, including 91.61% on paralinguistics. Self-hosted TTS used to win on cost. It is now winning on quality.

Read More: [Supertonic 3: 99M-Parameter On-Device TTS Across 31 Languages, No GPU Required](#)

Sources:

- [Fish Audio Open-Sources S2: Fine-Grained Control Meets Production Streaming \(Fish Audio\)](#)
- [Fish Audio S2 Technical Report \(arXiv\)](#)
- [Fish Audio S2 Pro model weights \(Hugging Face\)](#)
- [Fish Speech repository \(GitHub\)](#)
- [Audio Turing Test paper \(arXiv\)](#)

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